

Roof Lifecycle Cost Analysis

Planned maintenance is projected to save the owner of a typical Laurel school **\$1.87M** over the next 20 years.

That is 61.9% less than the reactive cost, or \$812K in today's dollars. It compares PaxSeal planned maintenance with fixing problems only after they leak, using Laurel costs and storm history.

MARKET	PROPERTY	ROOF AREA	SYSTEM	AGE	NEW ROOF TODAY
Laurel	K-12	120,000 sq ft	Mod bit	8 yrs Good	\$960K

PAXSEAL NET COST

\$1.15M

\$2.22M spent, less \$1.06M roof life left

REACTIVE NET COST

\$3.03M

\$3.19M spent, less \$157K roof life left

EMERGENCY COSTS CUT

\$1.6M

Unbudgeted repairs and storm damage avoided; \$4.25 back per fee dollar

NEW ROOF PRICE

2× in 15 yrs

\$960K today, \$2.36M by year 20

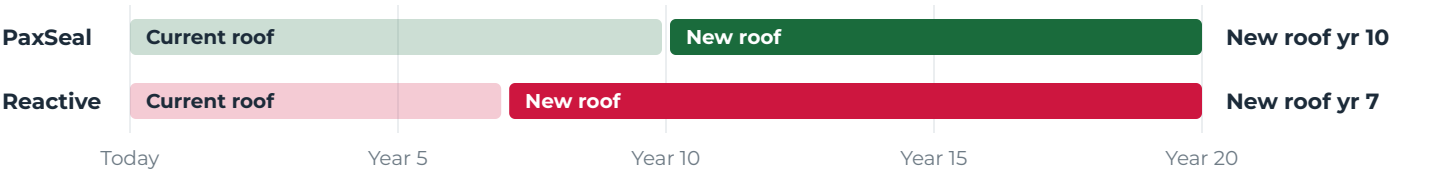
Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.



Roof life

When each scenario needs a new roof.



Where the savings come from

20-year totals

	PAXSEAL	REACTIVE	SAVINGS
PaxSeal program fees	\$441K	—	-\$441K
Repairs paid by owner	\$231K	\$1.81M	+\$1.58M
Expected storm damage	\$40K	\$60K	+\$20K
Roof replacements	\$1.51M	\$1.32M	-\$190K
Credit for roof life left	-\$1.06M	-\$157K	+\$905K
Net 20-year cost	\$1.15M	\$3.03M	+\$1.87M

What this means

- 1 **A later replacement.** Maintenance moves the next new roof from year 7 to year 10, so fewer roofs are bought over the period.
- 2 **Smaller repairs.** Problems found on a scheduled visit avoid emergency work, which costs about 3× as much. That saves \$1.58M in repairs.
- 3 **A predictable budget line.** PaxSeal turns unplanned emergency repairs into a fixed annual cost the board can plan for, and keeps leaks from closing classrooms.
- 4 **The program pays for itself.** Each \$1 of PaxSeal fees returns about \$4.25 in avoided cost.

Local risk profile

Prince George's County, MD (24033)

Thunderstorm wind / derecho, snow load

~3.5 major regional roof-damaging events per 20 years (2010 snow, 2012 derecho, 2016 blizzard). Expected storm damage to this roof over 20 years is **\$40K** if maintained and **\$60K** if neglected. A documented maintenance history also helps when an insurer questions a claim.

Hail	Moderate	■■■■■
Strong wind	High	■■■■■
Tornado	High	■■■■■
Hurricane	Moderate	■■■■■
Winter weather	Very High	■■■■■
Ice storm	Moderate	■■■■■

FEMA National Risk Index rating

Key assumptions

Confidence reflects source strength. Values will be refined as research continues.

ASSUMPTION	VALUE	BASIS	CONFIDENCE
Roof life	20 / 15 yrs	Maintained / reactive. Fannie Mae service-life tables; +5 yrs from NRC Canada roof study and manufacturer warranty terms	MEDIUM
Roof cost growth	4.6%/yr	BLS Producer Price Index, nonresidential roofing contractors, 2007–2025	HIGH
Replacement cost	\$8.00/sq ft	Laurel contractor price ranges	LOW
Reactive repair cost	3× planned	Industry estimates; no primary study found	LOW
Storm damage	0.13%/yr	Of roof cost. FEMA NRI expected annual building loss from hail, wind, tornado, hurricane and winter weather	MEDIUM
PaxSeal fee	\$16,400/yr	PAX program estimate for this roof size. Your proposal will state the actual fee	ESTIMATE
Leak consequences	1.30× warehouse	Scales reactive repair costs. Order supported by school-closure, hospital infection-control and insurer water-damage evidence; exact values are estimates	LOW

Estimates, not a quote. Projections use Laurel market research and industry data, in nominal dollars; actual costs will differ. The PaxSeal fee is an estimate and your proposal will state the actual price. Net cost credits the roof life remaining at year 20.